

# Multivariate Optimisation

## LLE – Mathematics and Statistics Skills

1. A company has the following total revenue ( $TR$ ) and total cost ( $TC$ ) functions to produce  $Q$  units of a good.

$$TR = 80Q$$

$$TC = 100 + Q^2$$

- (a) Write down the formula for the profit the company makes on selling  $Q$  units of this good.
  - (b) Find the quantity of goods that maximise the profit.
  - (c) Find the value of the maximum profit.
  - (d) Confirm that this quantity represents a maximum profit.
2. Consider the following function  $z$  of  $x$  and  $y$

$$z(x, y) = x^2 + y^2$$

- (a) Find the partial derivatives
  - i.  $\frac{\partial z}{\partial x}$
  - ii.  $\frac{\partial z}{\partial y}$
- (b) Solve the first order conditions,  $\frac{\partial z}{\partial x} = 0$ ,  $\frac{\partial z}{\partial y} = 0$ , to find the coordinates  $(x, y, z)$  of any local minimums, maximums, or saddle points.
- (c) At the coordinates of any points, found in part b, calculate the values of
  - i.  $\frac{\partial^2 z}{\partial x^2}$
  - ii.  $\frac{\partial^2 z}{\partial y^2}$
  - iii.  $\frac{\partial^2 z}{\partial x^2} \frac{\partial^2 z}{\partial y^2} - \left( \frac{\partial^2 z}{\partial x \partial y} \right)^2$

- (d) Determine whether points found are maximums, minimums, or saddle points.

3. Repeat the operations of question 2 for the following functions

(a)  $z = x^2 - y^2 + 20$

(b)  $z = 20x - x^2 + 6y - y^2$

(c)  $z = 2x^2 + 4x - 2xy + y^2$

4. Given the following function

$$z = x^2y - 2xy^2 + 3xy - 1$$

- (a) Show that the first order conditions can be written as

$$\begin{aligned}\frac{\partial z}{\partial x} &= y(2x - 2y + 3) = 0 \\ \frac{\partial z}{\partial y} &= x(x - 4y + 3) = 0\end{aligned}$$

- (b) Given that both first order conditions need to be satisfied simultaneously, find the 4 possible coordinates. Find the value of the function at these points.

5. A company producing two goods has an average cost  $AC$  function of

$$AC = Q_1^2 + 2Q_2^2 - 8Q_1 - 12Q_2 + 100$$

Find the quantities that minimise average cost, the average cost at this value, and confirm that this solution is a minimum.